DSA03-Natural Language Processing

List of Lab Program

2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we create a simple automaton to match strings ending with ;ab&; using python.

PROGRAM

```
def fsa(s):
    state = 0

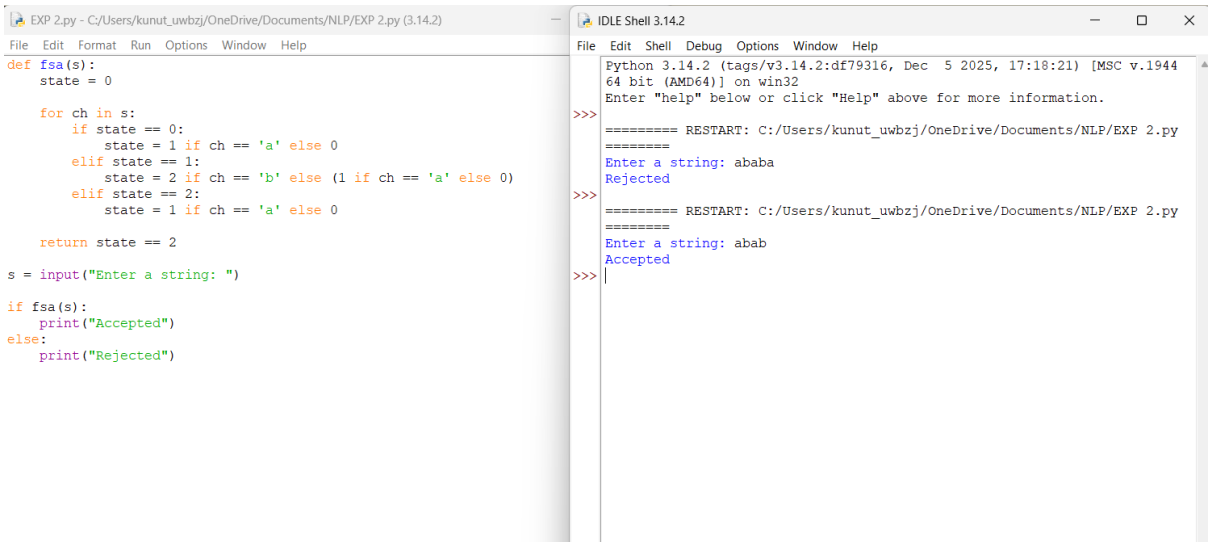
    for ch in s:
        if state == 0:
            state = 1 if ch == 'a' else 0
        elif state == 1:
            state = 2 if ch == 'b' else (1 if ch == 'a' else 0)
        elif state == 2:
            state = 1 if ch == 'a' else 0

    return state == 2

s = input("Enter a string: ")

if fsa(s):
    print("Accepted")
else:
    print("Rejected")
```

OUTPUT


EXP 2.py - C:/Users/kunut_uwbzj/OneDrive/Documents/NLP/EXP 2.py (3.14.2)
File Edit Format Run Options Window Help
def fsa(s):
state = 0
for ch in s:
if state == 0:
state = 1 if ch == 'a' else 0
elif state == 1:
state = 2 if ch == 'b' else (1 if ch == 'a' else 0)
elif state == 2:
state = 1 if ch == 'a' else 0
return state == 2
s = input("Enter a string: ")
if fsa(s):
print("Accepted")
else:
print("Rejected")
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944
64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: C:/Users/kunut_uwbzj/OneDrive/Documents/NLP/EXP 2.py
========
Enter a string: ababa
Rejected
>>>
========= RESTART: C:/Users/kunut_uwbzj/OneDrive/Documents/NLP/EXP 2.py
========
Enter a string: abab
Accepted
>>>